

homework 6

Question 1

A botany lab is testing two fertilizer treatments (trt1 and trt2) on plant growth. Using the PlantGrowth dataset (preloaded in R), complete the analysis in the following two parts: (1) Check statistical assumptions (normality and variance homogeneity); (2) analyze whether the treatments produce significantly different mean weights ($\alpha = 0.05$).

Question 2

A research team is investigating the effects of different anti-tumor drugs on the proliferation ability of lung cancer cells. In the experiment, “cell viability” (unit in “% of control”) was used as the evaluation indicator. The researchers set up four experimental groups, each with 6 biological replicates. The data are in the file “homework6-Q2.csv”.

The researchers first compared whether there were differences among the different drug treatment groups, and then further examined the combined effects of drug type and treatment time on cell viability.

- (1) Please use ANOVA to evaluate whether there are differences among the different drug treatments. If the ANOVA result is significant, please compare which groups have significant differences?
- (2) Please use Two-way ANOVA to analyze: Does drug type significantly affect cell viability? Does treatment time significantly affect cell viability? If both drug type and treatment time have significant effects on cell viability, please use `pairwise.t.test()` to compare the differences among all Drug-Time combinations, and use the BH method for multiple testing correction, and provide a brief summary.

Question 3

Using the built-in R dataset `swiss`, we will explore the relationships between socioeconomic factors and fertility, and build different regression models. Please answer the following three questions:

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1. Draw a scatterplot matrix for the variables Fertility, Education, Agriculture, and Infant.Mortality. Then, build a multiple linear regression model with Fertility as the dependent variable and the other three as independent variables, and interpret the model summary.
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2. Based on the significance of the variables ($\alpha = 0.05$) in the full linear model, remove the insignificant ones to build a simplified model. Compare the R^2 and Adjusted R^2 between the full and simplified models, and use ANOVA to test for significant differences between them.
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3. Create a binary variable high_fertility (1 = Fertility > median, 0 = otherwise). Build a logistic regression model using high_fertility as the dependent variable, with Education, Agriculture and Infant.Mortality as independent variables. Why do the effect size and significance differ between the two models (linear model vs. logistic model), even though they use the same variable?

Question 4

Using the R built-in dataset USArrests, complete the following analysis:

1. Use `prcomp()` to perform PCA on USArrests. Calculate the proportion of variance explained by each principal component. Report the cumulative variance explained by PC1 and PC2. If the cumulative variance explained should exceed 90%, determine the minimum number of principal components to retain.
2. Use `biplot()` to visualize the PCA result. Based on the direction and length of the variable arrows, identify which variables may be positively correlated and which variables contribute more strongly to the PC1-PC2 plane. Briefly interpret what PC1 and PC2 may represent.
3. Extract the scores of PC1 and PC2, and perform K-means clustering on the 50 states with $K = 3$. Output the cluster assignment for each state, and briefly describe the crime characteristics represented by different clusters.
4. Plot the states in the PC1-PC2 space and color them by K-means cluster. Comment on whether the clusters are clearly separated or partially overlapping. Finally, briefly evaluate the advantages and limitations of the “PCA dimensionality reduction + K-means clustering” workflow.

Question 5 (多组学数据分析的实战应用)

数据: “pbmc3k_filtered_gene_bc_matrices.tar.gz”

- 下载链接:
https://cf.10xgenomics.com/samples/cell/pbmc3k/pbmc3k_filtered_gene_bc_matrices.tar.gz
- 文件体积 7.26 MB, 解压后的文件结构如下:

```
└─filtered_gene_bc_matrices
  └─hg19
    barcodes.tsv
    genes.tsv
    matrix.mtx
```

任务:

1. 在 R 中安装 Seurat
2. 解压文件并运行下面的代码, 记录输出。提交作业时需包含代码运行结果
3. 回答代码注释中的问题

```
library(Seurat)
library(dplyr)
library(patchwork)
```

解压文件以后, 把“filtered_gene_bc_matrices”文件夹和当前代码放在一起

```
pbmc.data <- Read10X(data.dir = './filtered_gene_bc_matrices/hg19/')
pbmc <- CreateSeuratObject(
  counts = pbmc.data,
  project = "pbmc3k",
  min.cells = 3,
  min.features = 200
)
```

问题 1: 数据通常包含三个文件 (barcodes, genes, matrix), 请解释这三个文件分别代表什么?

```
pbmc[['percent.mt']] <- PercentageFeatureSet(pbmc, pattern = "^MT-") # 查找线粒体基因
pbmc <- subset(pbmc,
  subset = nFeature_RNA > 200 &
  nFeature_RNA < 2500 &
  percent.mt < 5 )
```

问题2：请解释上述代码中，每个过滤条件的目的。除了上面的三个过滤条件外，还有其他过滤条件吗，这些过滤条件的目的分别是什么？

```
pbmc <- NormalizeData(pbmc,  
  normalization.method = "LogNormalize",  
  scale.factor = 10000)
```

问题3：为什么需要对单细胞 UMI count 进行标准化？LogNormalize 的基本思想又是什么？

```
pbmc <- FindVariableFeatures(pbmc,  
  selection.method = "vst",  
  nfeatures = 2000)  
all.genes <- rownames(pbmc)  
pbmc <- ScaleData(pbmc, features = all.genes)  
pbmc <- RunPCA(pbmc,  
  features = VariableFeatures(object = pbmc))  
ElbowPlot(pbmc, ndims = 25)
```

问题4：PCA 的目的是什么？PC1、PC2、PC3 分别代表什么？结合 ElbowPlot，如何判断使用多少个 PC 进行后续分析？

```
pbmc <- FindNeighbors(pbmc, dims = 1:20)  
pbmc <- FindClusters( pbmc, resolution = 0.5)  
pbmc <- RunUMAP(pbmc, dims = 1:20)  
DimPlot(pbmc,  
  reduction = "umap",  
  label = TRUE  
)
```

问题5：UMAP 是一种非线性降维算法，其主要作用是将 PCA 后的高维细胞表达数据映射到二维或三维空间，同时尽可能保留细胞之间的邻近关系。在 UMAP 图中，如果两个 cluster 分得很开，可能说明什么？

```
cluster_markers <- FindAllMarkers(  
  pbmc,  
  only.pos = TRUE,  
  min.pct = 0.25,  
  logfc.threshold = 0.25  
)  
library(dplyr)  
top10 <- cluster_markers %>% group_by(cluster) %>% top_n(n = 10, wt = avg_log2FC)  
head(top10)
```

问题6：FindAllMarkers 步骤通过比较基因表达量寻找每个 cluster 的标记基因，其中涉及了假设检验的步骤。在 Seurat 中，默认的检验方法是什么？为什么 Seurat 选择这种方法进行假设检验？